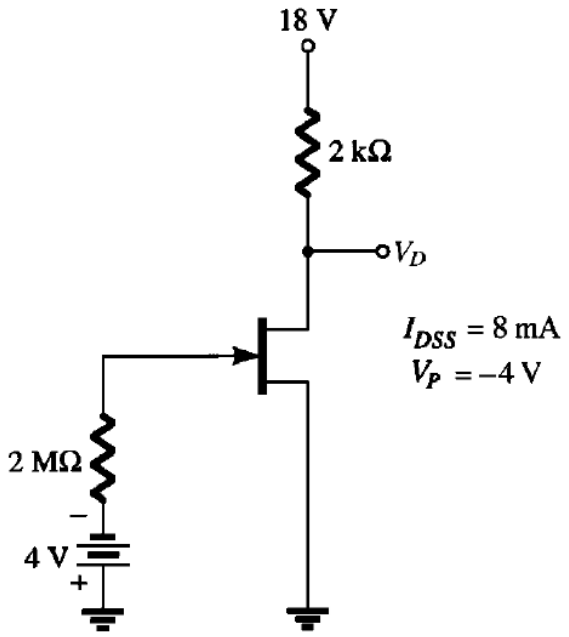
FET DC bias

Practice questions · Day 17 · Mon 9 Nov · pack E10

5. Practice

From your course files

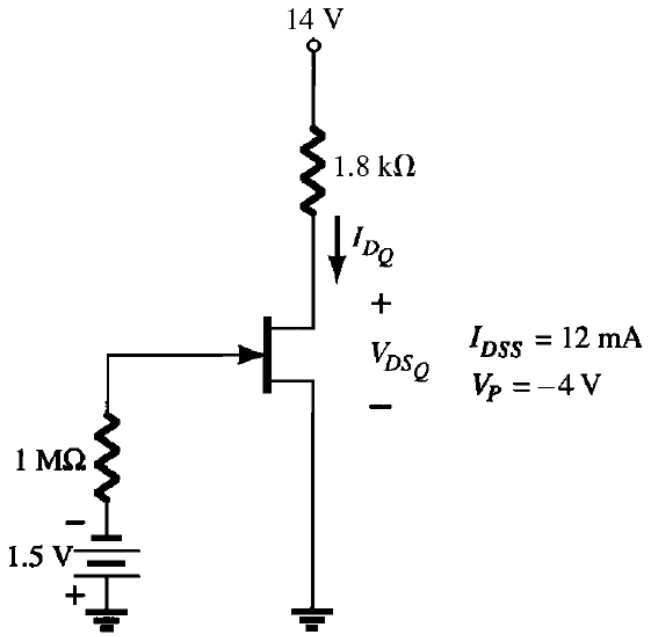
WS8 Q2 (course question). Determine V_D and V_{GS} for the fixed-bias configuration of Fig. 2.



WS8 Fig. 2 (from the workshop sheet)

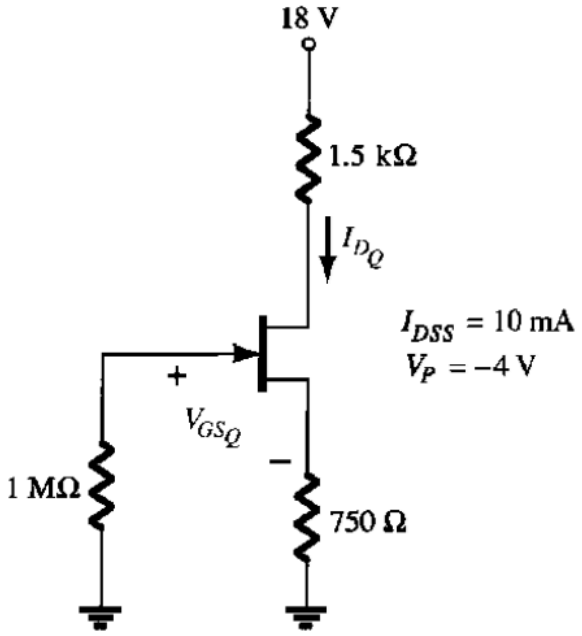
WS8 Q1 (course question). For the fixed-bias configuration of Fig. 1:

a. Sketch the transfer characteristics of the device.
b. Determine I_{D_Q} and V_{DS_Q}.
c. Using Shockley's equation, solve for I_{D_Q} and then find V_{DS_Q}. Compare with the solutions of part (b).



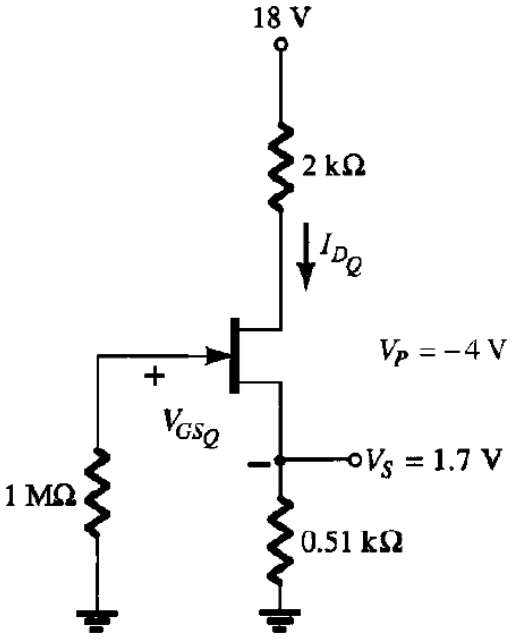
WS8 Fig. 1 (from the workshop sheet)

WS8 Q3 (course question). Determine I_{D_Q} for the network of Fig. 3 using a purely mathematical approach.



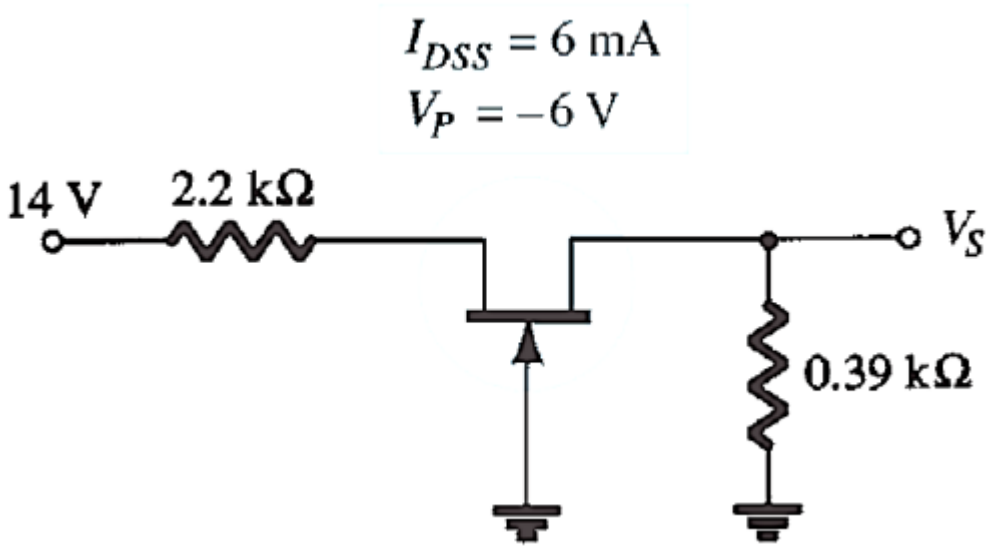
WS8 Fig. 3 (from the workshop sheet)

WS8 Q4 (course question). Given the measurement $V_S = 1.7$ V for the network of Fig. 4, determine: (a) I_{D_Q}, (b) V_{GS_Q}, (c) I_{DSS}, (d) V_D and (e) V_{DS}.



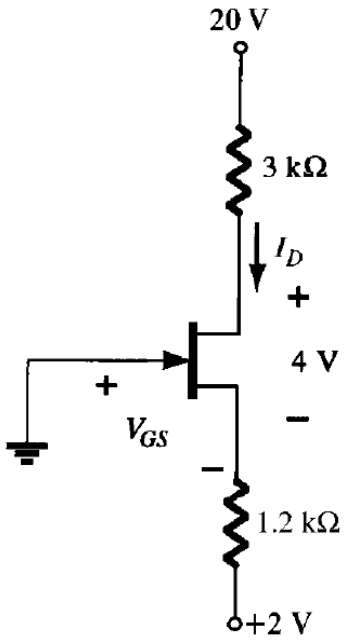
WS8 Fig. 4 (from the workshop sheet)

WS9 Q1 (course question). Find V_S for the network of Fig. 1.



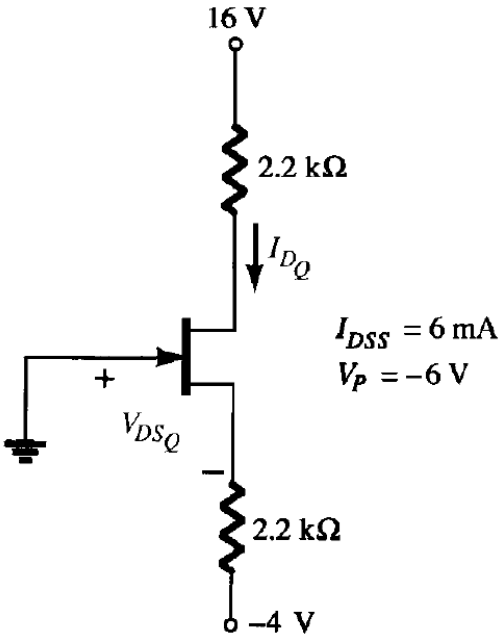
WS9 Fig. 1 (from the workshop sheet)

WS9 Q3 (course question). Given $V_{DS} = 4$ V for the network of Fig. 3, determine: I_D, V_D, V_S and V_{GS}.



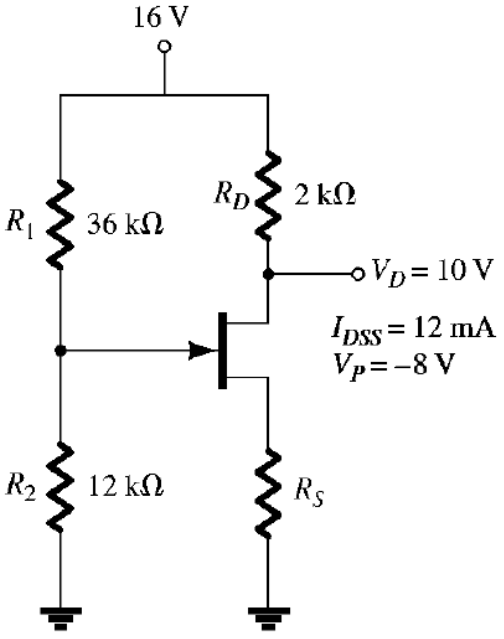
WS9 Fig. 3 (from the workshop sheet)

WS9 Q2 (course question). For the network of Fig. 2, determine: (a) I_{D_Q} and V_{GS_Q} and (b) V_{DS} and V_S.



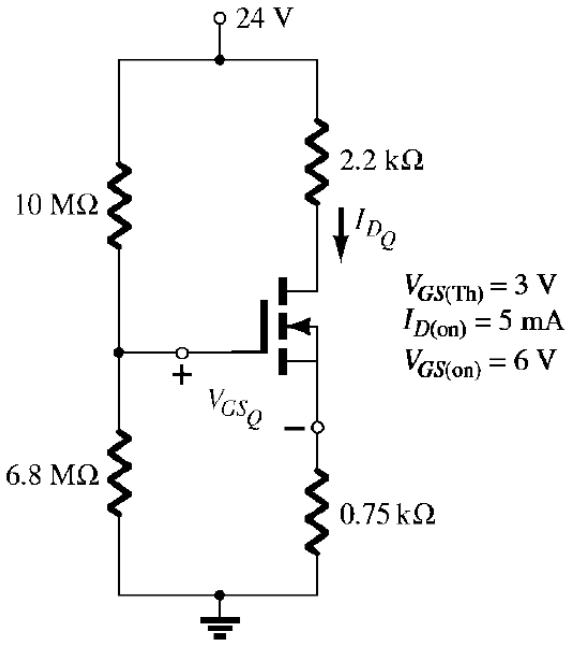
WS9 Fig. 2 (from the workshop sheet)

WS8 Q5 (course question). Determine the value of R_S the network of Fig. 5 to establish $V_D = 10$ V.



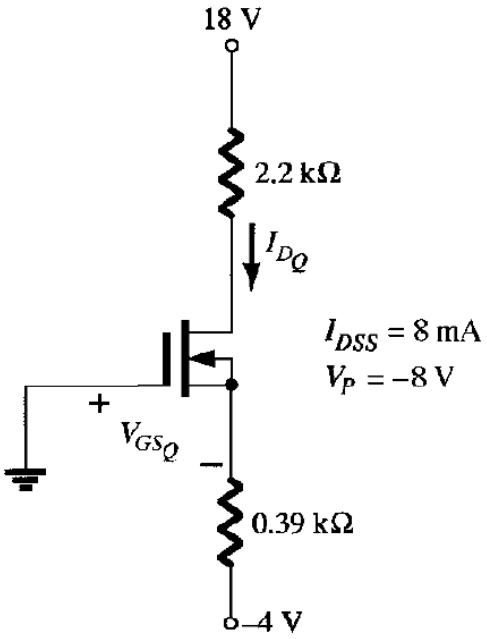
WS8 Fig. 5 (from the workshop sheet)

WS8 Q6 (course question). For the voltage-divider configuration of Fig. 6, determine: (a) I_{D_Q} and V_{GS_Q} and (b) V_D and V_S.



WS8 Fig. 6 (from the workshop sheet)

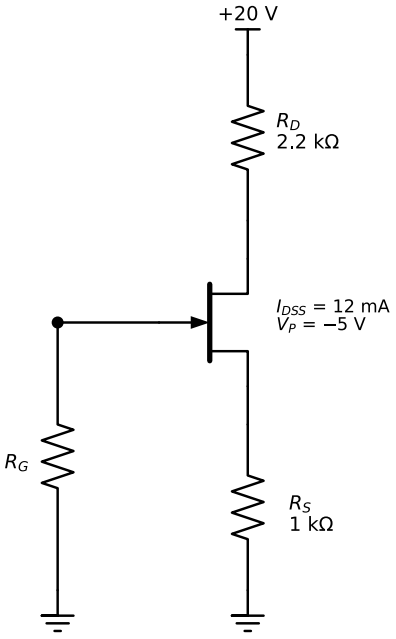
WS9 Q4 (course question). For the network of Fig. 4, determine: (a) I_{D_Q} and V_{GS_Q} and (b) V_{DS} and V_S.



WS9 Fig. 4 (from the workshop sheet)

New questions

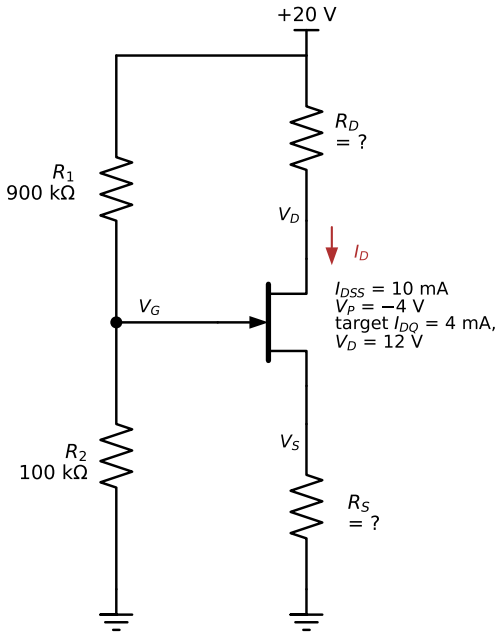
N1. [4 marks] A JFET with $I_{DSS} = 8$ mA and $V_P = -6$ V is fixed-biased with $V_{GG} = 2$ V (so $V_{GS} = -2$ V), $R_D = 1.5$ kΩ and $V_{DD} = 15$ V. Find I_{DQ} and V_{DSQ}, and confirm saturation.



Circuit for N2: self-bias JFET

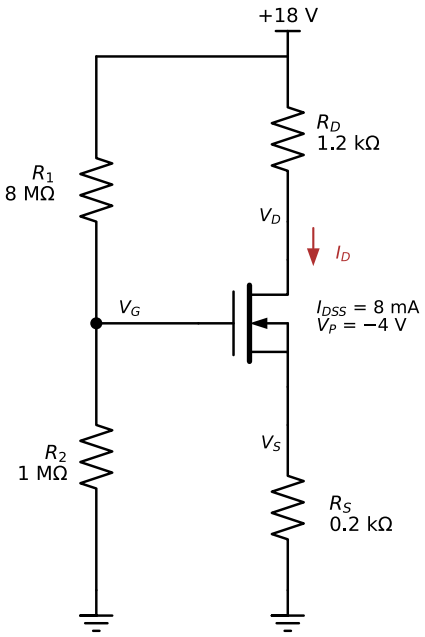
N2. [6 marks] A self-bias JFET has $I_{DSS} = 12$ mA, $V_P = -5$ V, $R_S = 1$ kΩ, $R_D = 2.2$ kΩ and $V_{DD} = 20$ V.

a. Write the bias line and solve for I_{DQ} and V_{GSQ}, justifying your choice of root. [4]
b. Find V_{DS} and check saturation. [2]



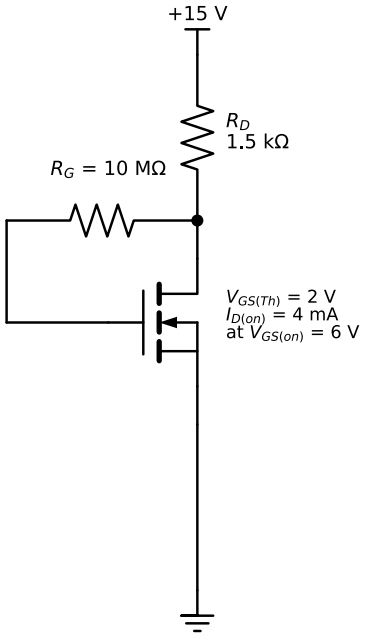
Circuit for N3: voltage-divider JFET, R_S and R_D to be designed

N3. [6 marks] Design. A JFET with $I_{DSS} = 10$ mA and $V_P = -4$ V uses voltage-divider bias with $V_{DD} = 20$ V, $R_1 = 900$ kΩ and $R_2 = 100$ kΩ. Choose R_S for $I_{DQ} = 4$ mA, and R_D for $V_D = 12$ V. Then check saturation.



Circuit for N4: D-MOSFET voltage-divider bias

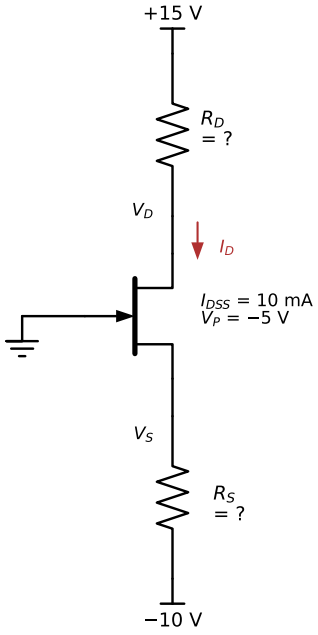
N4. [6 marks] An n-channel D-MOSFET with $I_{DSS} = 8$ mA and $V_P = -4$ V has $V_{DD} = 18$ V, $R_1 = 8$ MΩ, $R_2 = 1$ MΩ, $R_D = 1.2$ kΩ and $R_S = 0.2$ kΩ. Find I_{DQ}, V_{GSQ}, V_D, V_S and V_{DS}. Comment on the sign of V_{GSQ} and on saturation.



Circuit for N5: E-MOSFET drain-feedback bias

N5. [8 marks] Chained. An n-channel E-MOSFET has $V_{GS(\text{Th})} = 2$ V and $I_{D(\text{on})} = 4$ mA at $V_{GS(\text{on})} = 6$ V. It is used in drain-feedback bias with $V_{DD} = 15$ V, $R_D = 1.5$ kΩ and $R_G = 10$ MΩ.

a. Find k. [1]
b. Find I_{DQ} and V_{GSQ}. [4]
c. State V_{DSQ} and explain why no saturation check is needed. [1]
d. Find the new R_D that would give $V_{DSQ} = 8$ V. [2]



Circuit for N6: common-gate JFET with R_S returned to -10 V

N6. [8 marks] Common-gate design, hard. A JFET with $I_{DSS} = 10$ mA and $V_P = -5$ V has its gate grounded. R_D goes to $V_{DD} = +15$ V, and R_S goes to $-V_{SS} = -10$ V.

a. Find R_S for $I_{DQ} = 5$ mA. [3]
b. Find R_D for $V_D = 6$ V, then find V_S and V_{DS} and check saturation. [3]
c. Someone fits $R_D = 2.2$ kΩ instead. Is the JFET still saturated? What is the largest R_D that keeps it saturated, assuming I_D stays at 5 mA? [2]

6. Quick recall check

1. Write the self-bias line for a JFET and say where it crosses the axes.
2. For a JFET quadratic, which root do you keep?
3. Can I_{DQ} exceed I_{DSS}? For which device, and when?
4. State the saturation condition for a JFET and for an E-MOSFET.
5. For an E-MOSFET with drain feedback, what is the relation between V_{DS} and V_{GS}, and why?